Some counterexamples on higher order Bergman kernels

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Abstract

We present three counterexamples in the theory of higher order Bergman kernels. The first answers in the negative a convexity question of Błocki–Zwonek; the second concerns the equality case of the higher order Suita conjecture; and the third shows that a natural identity between higher order p-Bergman kernels and ξ-p-Bergman kernels fails for $0 < p < 1$. The constructions and proofs of these counterexamples come from AI tools.

1 Introduction

Let $D \subset \mathbb{C}^n$ be a bounded pseudoconvex domain and let $w \in D$. The Bergman kernel of D at w is defined by

$$K_D(w) := \sup\Big\{|f(w)|^2 : \ f \in \mathcal{O}(D), \ \|f\|_D := \Big(\int_D |f|^2 \, d\lambda\Big)^{1/2} \le 1\Big\},$$

where $d\lambda$ denotes the Lebesgue measure. Following [4], for a homogeneous polynomial H of degree k, $H(z) = \sum_{|\alpha|=k} a_\alpha z^\alpha$, we write $P_H(f) := \sum_{|\alpha|=k} a_\alpha D^\alpha f$, and we define the higher order Bergman kernel

$$K_D^H(z) := \sup\Big\{|P_H(f)(z)|^2 : \ f \in \mathcal{O}(D), \ f^{(j)}(z) = 0 \ (0 \le j < k), \ \|f\|_D \le 1\Big\}.$$

In dimension one, $H(z) = z^k$ and $K_D^{z^k}(z)$ will also be denoted by $K_D^{(k)}(z)$.

Let $G_D(\cdot, w)$ be the (pluricomplex) Green function of D with pole at w, and assume $w = 0$. For $a \le 0$ we denote

$$D_a := e^{-a}\big\{G_D(\cdot, 0) < a\big\}.$$

The main theorem of [4] asserts that $a \mapsto K_{D_a}^H(0)$ is non-decreasing, and it is noted that

$$K_{D_a}^H(0) = e^{2(n+k)a} K_{\{G_D(\cdot,0)<a\}}^H(0).$$

The three counterexamples in this paper concern three natural extremal questions for these kernels.

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1.1 The convexity question of Błocki–Zwonek

The main theorem of [4] also motivates the following question raised in [4, Remark 3].

Question 1.1 (Błocki–Zwonek, [4, Remark 3]). *Is the function*

$$(-\infty, 0] \ni a \longmapsto \log K^H_{D_a}(w)$$

convex, as it is in the case $H \equiv 1$?

The convexity in the case $H \equiv 1$ follows from Berndtsson's theorem on the plurisubharmonic variation of Bergman kernels (see the final remark in [2]). Our first result shows that Question 1.1 has a negative answer in general.

theorem 1.2. *For $Q > 1$ sufficiently large set $c := 2Q$ and*

$$D_c := \left\{ z \in \mathbb{C} : \; |z|^2 < |1 - cz^2| \right\}.$$

Then D_c is a domain in $\mathbb{C}$ (indeed $\mathbb{C}$ minus two disjoint Jordan regions, hence unbounded and doubly connected), $0 \in D_c$, and with $H(z) = z^6$ the function

$$(-\infty, 0] \ni a \longmapsto \log K^{z^6}_{(D_c)_a}(0)$$

is <u>not</u> convex. More precisely, for

$$a_- := -\log 2, \qquad a_0 := -\tfrac{1}{2}\log 2, \qquad a_+ := 0,$$

so that $a_0 = (a_- + a_+)/2$, we have

$$\log K^{z^6}_{(D_c)_{a_0}}(0) > \frac{1}{2}\Big(\log K^{z^6}_{(D_c)_{a_-}}(0) + \log K^{z^6}_{(D_c)_{a_+}}(0) \Big),$$

and the difference is

$$\frac{3}{64Q^2} + O(Q^{-4}) \qquad (Q \to \infty).$$

The proof of Theorem 1.2 is divided into three steps, each of independent interest:

1. an explicit formula for the Green function of D_c and the observation that the scaled domains $(D_c)_a$ are again of the same form (Section 2.1);
2. an exact reduction of the extremal problem defining $K^{z^6}_{D_q}(0)$ to the third Taylor coefficient functional on a weighted Bergman space of the unit disk (Section 2.2);
3. a rigorous asymptotic expansion of the associated infinite Gram matrix, valid up to order q^{-4}, which yields the precise size of the convexity defect (Section 2.3).

1.2 Higher order Suita's conjecture

Let Ω be a planar domain in $\mathbb{C}$ bounded by n analytic Jordan curves: $\Gamma_1, \ldots, \Gamma_n$, where we set Γ_n to be the boundary of the unbounded component of $\mathbb{C} \setminus \Omega$. According to the solvability of the Dirichlet problem on planar domains, for each $j \in \{1, \ldots, n-1\}$ there exists a harmonic function u_j on $\overline{\Omega}$ such that

$$u_j|_{\Gamma_j} \equiv 1, \qquad u_j|_{\Gamma_k} \equiv 0 \quad \forall k \in \{1, \ldots, n\} \setminus \{j\}.$$

We recall the higher order Suita conjecture in the form in which it is used below. The classical Suita conjecture asserts that for a planar domain Ω with Green function G_Ω and $w \in \Omega$,

$$K_\Omega(w) \geq \frac{c_\Omega(w)^2}{\pi},$$

where

$$c_\Omega(w) := \exp\Big(\lim_{z\to w}\big(G_\Omega(z,w) - \log|z-w|\big)\Big)$$

is the logarithmic capacity of $\mathbb{C} \setminus \Omega$ with respect to w; for the unit disk the two sides coincide. Its m-th order analogue, proved in [3], states that for every $m \geq 0$,

$$K_\Omega^{(m)}(w) \geq \frac{m!(m+1)!\, c_\Omega(w)^{2m+2}}{\pi},$$

with equality again for the unit disk. In higher dimensions, Błocki–Zwonek [4] proved the general inequality $K_D^H(z) \geq K_{I_D(z)}^H(0)$ for every homogeneous polynomial H, where $I_D(z)$ denotes the Azukawa indicatrix of D at z; in dimension one, for $H(z) = z^m$, it is exactly the inequality above.

Guan–Sun–Yuan [7] proved a weighted version of the Suita conjecture for higher derivatives, and their result yields the following characterization of the equality locus: for a planar domain bounded by n analytic Jordan curves, the equality in the m-th order Suita conjecture holds at z if and only if

$$(m+1)u_j(z) \in \mathbb{Z}, \qquad j = 1, \dots, n-1.$$

Thus, if all $u_j(z)$ are rational, then choosing $m+1$ to be a common denominator of the $u_j(z)$ gives equality at z; conversely any such equality forces $u_j(z) \in \mathbb{Q}$ for all j. Hence a domain in which, at every point, some triple $(u_1(z), u_2(z), u_3(z))$ fails to be rational has the property that equality in the higher order Suita conjecture never holds.

Recently, in the Annales de l'Institut Fourier, Xu–Zhou [8] constructed, for every $n \geq 3$ and every sufficiently large integer M, a family of smoothly bounded n-connected planar domains on which the equality in the m-th order Suita conjecture fails at every point for each $m = 0, 1, \dots, M$. Our next result complements and strengthens this: it exhibits a four-connected domain on which the equality in the higher order Suita conjecture fails at every point and for every order $m \geq 0$.

THEOREM 1.3. *There exists a planar domain Ω in $\mathbb{C}$ bounded by 4 analytic Jordan curves, such that for the corresponding harmonic functions u_j, $j = 1, 2, 3$, there does not exist $z \in \Omega$ with*

$$u_j(z) \in \mathbb{Q}, \qquad j = 1, 2, 3.$$

In particular, at no point of Ω does the equality in the higher order Suita conjecture hold.

We prove Theorem 1.3 by a Baire-category (generic parameter) construction. The same construction works for every connectivity $n \geq 4$; we record this generalization in Subsection 3.1.

1.3 The ξ-reduction fails for $0 < p < 1$

For $p > 0$ and a homogeneous polynomial H of degree k, the higher order p-Bergman kernel $K_D^{H,p}(z)$ and, for a functional $\xi \in \ell_1^{(n)}$, the L^p ξ-Bergman kernel $K_{\xi,D,p}(z)$ are defined in Section 4. Bao–Guan–Sun [9] proved that for every $p \in [1, +\infty)$,

$$K_D^{H,p}(z) = \inf_{\xi \in S_H} K_{\xi,D,p}(z),$$

where S_H is the affine family of functionals whose degree-k part is prescribed by H; see Theorem 1.7 in [9].

It is natural to ask: whether the identity persists for $0 < p < 1$? Our third counterexample answers this question in the negative. Already in the simplest one-dimensional case the equality fails.

THEOREM 1.4. *Let $\mathbb{D} \subset \mathbb{C}$ be the unit disk, $p = 1/2$, $H(z) = z$ and $z = 0$. Then*

$$K_{\mathbb{D}}^{H,1/2}(0) = \frac{5}{4\pi},$$

while

$$\inf_{\xi \in S_H} K_{\xi,\mathbb{D},1/2}(0) > \frac{5}{4\pi}.$$

Consequently the identity

$$K_{\mathbb{D}}^{H,1/2}(0) = \inf_{\xi \in S_H} K_{\xi,\mathbb{D},1/2}(0)$$

fails, and the infimum is not attained at the common value.

2 Non-convexity of the logarithm of a higher order Bergman kernel

The proof of Theorem 1.2 is divided into three steps, which are treated in the next three subsections.

2.1 The Green function of D_c and the scaling law

Throughout this section we write $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$.

PROPOSITION 2.1. *Let $c > 0$ and*

$$F_c(z) := \frac{z^2}{1 - cz^2}.$$

Then the domain $D_c = \{|F_c| < 1\}$ satisfies $0 \in D_c$, and the mapping

$$F_c : D_c \longrightarrow \mathbb{D} \setminus \{-1/c\}$$

is proper of degree 2, vanishing to order 2 at 0. Consequently the (negative) pluricomplex Green function of D_c with pole at 0 is

$$G_{D_c}(z, 0) = \frac{1}{2} \log |F_c(z)| = \log |z| - \frac{1}{2} \log |1 - cz^2|.$$

Proof. The identity $D_c = \{|F_c| < 1\}$ is immediate from the definitions. The rational function F_c is holomorphic on D_c: the poles $\pm c^{-1/2}$ are not contained in D_c, since $|F_c| \to +\infty$ near them. The equation $F_c(z) = \xi$ reads $z^2 = \xi(1 - cz^2)$, i.e. $z^2 = \xi/(1 + c\xi)$, which has two solutions for every $\xi \notin \{-1/c, \infty\}$, so F_c is proper of degree 2 and $F_c(0) = 0$ with $F_c'(0) = 0$, $F_c''(0) = 2 \neq 0$.

For a proper holomorphic map of degree m between planar domains with $f(0) = 0$, the Green functions satisfy $G_D(z, 0) = \frac{1}{m} G_{D'}(f(z), 0)$. Since $\mathbb{D} \setminus \{-1/c\}$ is obtained from $\mathbb{D}$ by removing a polar set, its Green function with pole 0 is $\log |\cdot|$. Taking $m = 2$ gives the formula. ■

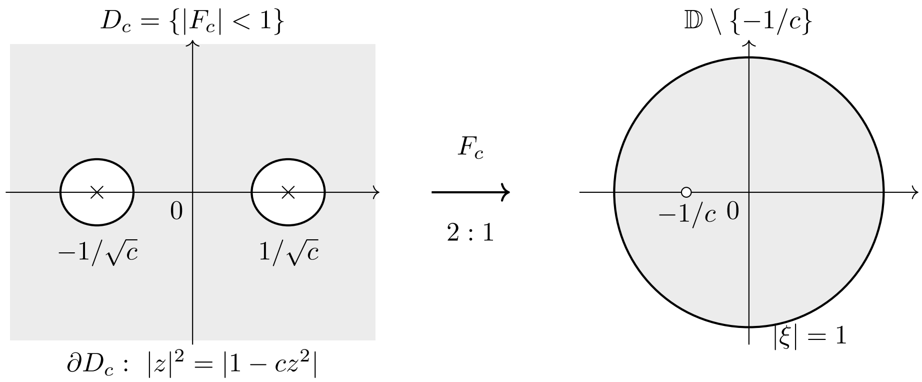

Figure 1: The domain $D_c = \{z \in \mathbb{C} : |z|^2 < |1 - cz^2|\}$ in the z-plane (shaded; the two holes are centered at $\pm 1/\sqrt{c}$ and the boundary is $|z|^2 = |1 - cz^2|$) and its image $F_c(D_c) = \mathbb{D} \setminus \{-1/c\}$ in the ξ-plane. The map F_c is proper of degree 2 with $F_c(0) = 0$ (a double zero) and $F_c(\infty) = -1/c$.

For $a \le 0$ we use the notation of [4]:

$$(D_c)_a := e^{-a}\big\{G_{D_c}(\cdot, 0) < a\big\}.$$

Proposition 2.2. *For $a \le 0$ and $q := ce^{2a}$ we have*

$$(D_c)_a = D_q := \big\{x \in \mathbb{C} :\ |x|^2 < |1 - qx^2|\big\}.$$

Proof. By Proposition 2.1, $G_{D_c}(z, 0) < a$ is equivalent to $|F_c(z)| < e^{2a}$. Writing $z = e^a x$, we have

$$F_c(e^a x) = e^{2a} \frac{x^2}{1 - ce^{2a}x^2} = e^{2a} F_q(x),$$

with $q = ce^{2a}$. Hence $|F_c(e^a x)| < e^{2a}$ is equivalent to $|F_q(x)| < 1$, i.e. $x \in D_q$. Since $(D_c)_a = \{x : e^a x \in \{G_{D_c}(\cdot, 0) < a\}\}$, the claim follows. ∎

Since $\log q = \log c + 2a$ is affine in a, the convexity of $a \mapsto \log K^{z^6}_{(D_c)_a}(0)$ is equivalent to the convexity of $s \mapsto \log K^{z^6}_{D_{e^s}}(0)$ in $s = \log q$. We shall therefore study the family D_q, $q > 1$.

2.2 Reduction to a weighted Bergman kernel on the unit disk

For $q > 1$ we abbreviate $K_q := K^{z^6}_{D_q}(0)$. Since D_q is invariant under $x \mapsto -x$, the sixth derivative at 0 only sees the even part of a holomorphic function, and we may restrict to

$$f(x) = g(x^2), \qquad g \in \mathcal{O}(U_q),$$

where

$$U_q := \big\{y \in \mathbb{C} :\ |y| < |1 - qy|\big\} = \mathbb{C} \setminus \overline{B\Big(\frac{q}{q^2 - 1}, \frac{1}{q^2 - 1}\Big)}$$

is the image of D_q under the two-to-one map $y = x^2$.

We record the elementary facts used below.

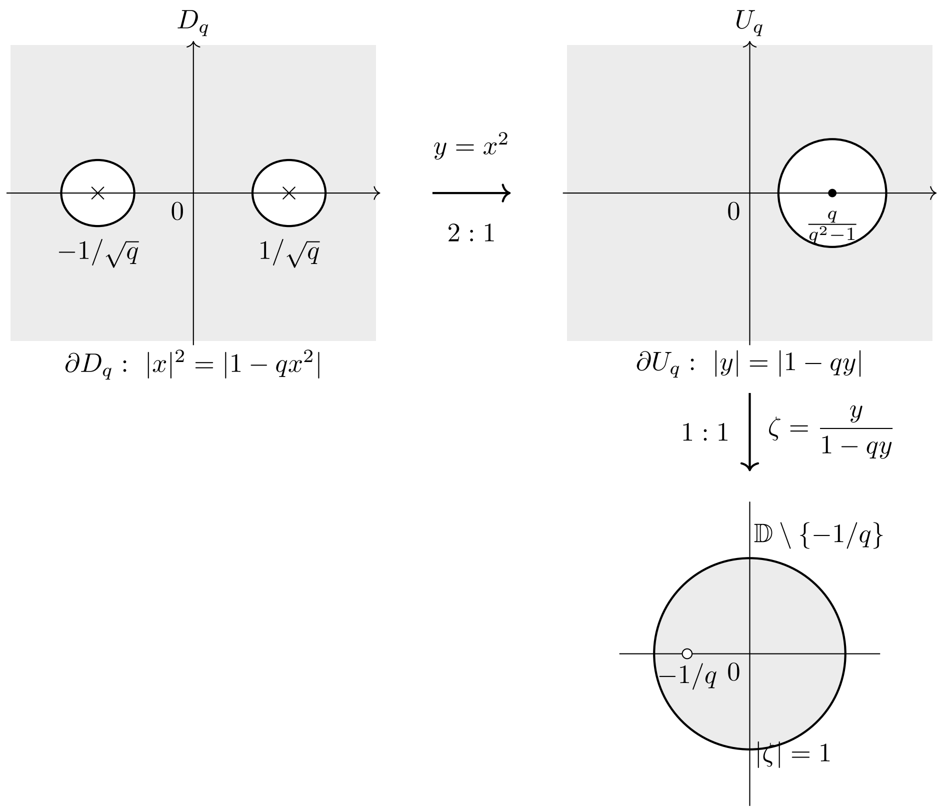

Figure 2: The domain $D_q = \{x \in \mathbb{C} : |x|^2 < |1 - qx^2|\}$ in the x-plane (shaded; the two holes are centered at $\pm\sqrt{q/(q^2-1)}$ and contain the poles $\pm 1/\sqrt{q}$), its image $U_q = \{y \in \mathbb{C} : |y| < |1 - qy|\} = \mathbb{C} \setminus \overline{B\big(q/(q^2-1), 1/(q^2-1)\big)}$ in the y-plane under the two-to-one map $y = x^2$, and the unit disk $\mathbb{D} \setminus \{-1/q\}$ in the ζ-plane obtained via the Möbius map $\zeta = y/(1 - qy)$ $(1:1)$.

Lemma 2.3. *With the notation above, the following hold.*

(i) For $f(x) = g(x^2)$,

$$\|f\|^2_{L^2(D_q)} = \frac{1}{2}\int_{U_q} |g(y)|^2 \, \frac{dA(y)}{|y|}.$$

(ii) The conditions $f(0) = f'(0) = \cdots = f^{(5)}(0) = 0$ and $f^{(6)}(0) = 1$ are equivalent to

$$g(y) = b_3 y^3 + O(y^4) \quad (y \to 0), \qquad b_3 = \frac{1}{6!}.$$

(iii) Under the Möbius transformation

$$\zeta = \frac{y}{1-qy}, \qquad y = \frac{\zeta}{1+q\zeta},$$

which maps U_q biholomorphically onto $\mathbb{D} \setminus \{-1/q\}$, the square integrability of $|g|^2/|y|$ forces $g(y(\zeta))$ to vanish at $\zeta = -1/q$. Hence there exists $h \in \mathcal{O}(\mathbb{D})$ with

$$g(y(\zeta)) = (1+q\zeta)h(\zeta),$$

and

$$\|f\|^2_{L^2(D_q)} = \frac{1}{2}\int_{\mathbb{D}} |h(\zeta)|^2 \, \frac{dA(\zeta)}{|\zeta|\,|1+q\zeta|}.$$

(iv) In the variable ζ, the normalization $b_3 = 1/6!$ becomes

$$h(\zeta) = \frac{1}{6!}\zeta^3 + O(\zeta^4).$$

Proof. (i) The map $y = x^2$ is two-to-one from D_q onto U_q, and $dA(y) = 4|x|^2 dA(x) = 4|y|\, dA(x)$, whence $dA(x) = dA(y)/(4|y|)$.

(ii) Writing $g(y) = \sum_{m\geq 0} g_m y^m$, we have $f(x) = \sum_m g_m x^{2m}$; the first six derivatives vanish at 0 iff $g_0 = g_1 = g_2 = 0$, and $f^{(6)}(0) = 6!\, g_3$.

(iii) The Möbius map is a biholomorphism between U_q and $\mathbb{D} \setminus \{-1/q\}$ because $y = \zeta/(1+q\zeta)$ and

$$1 - qy = \frac{1}{1+q\zeta}, \qquad |y| < |1-qy| \iff |\zeta| < 1.$$

Near $\zeta = -1/q$ one has $y \sim -1/(q^2(\zeta + 1/q)) \to \infty$. Since g is holomorphic on U_q, including the point at infinity, $g(y(\zeta))$ is holomorphic near $\zeta = -1/q$; if it did not vanish there, then $|g|^2/|y| \sim |g(-1/q)|^2\,|1+q\zeta|^3$ would not be integrable near $\zeta = -1/q$, contradicting the finiteness of $\|f\|^2$. Hence $g(y(\zeta))$ vanishes at $\zeta = -1/q$, so the quotient $h = g(y(\zeta))/(1+q\zeta)$ is holomorphic on $\mathbb{D}$. The change of variables $y = \zeta/(1+q\zeta)$ has Jacobian $dA(y) = |1+q\zeta|^{-4} dA(\zeta)$, and $|y|^{-1} = |1+q\zeta|/|\zeta|$, so

$$\frac{1}{2}\int_{U_q} |g|^2 \frac{dA(y)}{|y|} = \frac{1}{2}\int_{\mathbb{D}} |h|^2 \frac{dA(\zeta)}{|\zeta|\,|1+q\zeta|}.$$

(iv) From $y = \zeta + O(\zeta^2)$ we get $g(y(\zeta)) = g_3\zeta^3 + O(\zeta^4)$; dividing by $(1+q\zeta)$ does not change the leading term. ■

We now introduce the infinite Gram matrix. For $i, j \geq 3$ set

$$M_{ij}(q) := \int_{\mathbb{D}} \frac{\zeta^i \bar{\zeta}^j}{|\zeta|\,|1+q\zeta|}\, dA(\zeta),$$

and let $K_3(q) := \big(M(q)^{-1}\big)_{33}$ be the $(3,3)$-entry of the inverse of the infinite matrix $M(q)$ (which is positive definite on the space of holomorphic functions h with $\int |h|^2/(|\zeta||1+q\zeta|)\, dA < \infty$).

Proposition 2.4. *For every $q > 1$,*

$$K_q = K^{z^6}_{D_q}(0) = 2(6!)^2\, K_3(q).$$

Proof. By definition,

$$K_q^{-1} = \min\Big\{ \|f\|^2_{L^2(D_q)} :\ f \in \mathcal{O}(D_q),\ f^{(j)}(0) = 0\ (0 \leq j < 6),\ f^{(6)}(0) = 1 \Big\}.$$

By Lemma 2.3 this equals

$$\min\Big\{ \frac{1}{2} \int_{\mathbb{D}} |h|^2 \frac{dA}{|\zeta|\,|1+q\zeta|} :\ h \in \mathcal{O}(\mathbb{D}),\ h(\zeta) = \frac{1}{6!}\zeta^3 + O(\zeta^4) \Big\}.$$

Writing $h(\zeta) = \frac{1}{6!}\zeta^3 + \sum_{m \geq 4} c_m \zeta^m$, the minimum over the higher coefficients is

$$\frac{1}{2(6!)^2} \big(M(q)^{-1}\big)_{33},$$

and taking the reciprocal gives the claim. ■

2.3 Asymptotics of the Gram matrix

Set $\varepsilon := 1/q$ and $A(\varepsilon) := qM(q)$, so that

$$A_{ij}(\varepsilon) = \int_{\mathbb{D}} \frac{\zeta^i \bar{\zeta}^j}{|\zeta|\,|\zeta+\varepsilon|}\, dA(\zeta), \qquad i, j \geq 3.$$

We write $D = \operatorname{diag}(\pi/3, \pi/4, \pi/5, \dots)$, and for the off-diagonal terms we use the convention that $B_{i,i+1} = B_{i+1,i}$ for $i \geq 3$.

Proposition 2.5. *The matrix function $A(\varepsilon)$ admits the expansion*

$$A(\varepsilon) = D + \varepsilon B + \varepsilon^2 C + \varepsilon^3 E + O(\varepsilon^4)$$

in the sense of bounded quadratic forms on the Hilbert space $\{h = \zeta^3 v :\ v \in L^2_h(\mathbb{D})\}$ with norm $\int |h|^2 |\zeta|^{-2} dA$. The nonzero leading terms are

$$D_{ii} = \frac{\pi}{i}, \qquad B_{i,i+1} = B_{i+1,i} = -\frac{\pi}{2i}, \qquad C_{ii} = \frac{\pi}{4(i-1)}, \qquad i \geq 3.$$

Consequently, for $\varepsilon > 0$ sufficiently small,

$$\big(A(\varepsilon)^{-1}\big)_{33} = \frac{3}{\pi} - \frac{1}{8\pi}\varepsilon^2 + O(\varepsilon^4).$$

Proof. The expansion of the kernel in the outer region $|\zeta| \geq 2\varepsilon$,

$$\frac{1}{|\zeta+\varepsilon|} = \frac{1}{|\zeta|}\Big(1+\frac{2\varepsilon\operatorname{Re}\zeta}{|\zeta|^2}+\frac{\varepsilon^2}{|\zeta|^2}\Big)^{-1/2} = \frac{1}{|\zeta|}-\frac{\varepsilon\operatorname{Re}\zeta}{|\zeta|^3}+\varepsilon^2\frac{3(\operatorname{Re}\zeta)^2-|\zeta|^2}{2|\zeta|^5}+O\Big(\frac{\varepsilon^3}{|\zeta|^4}\Big),$$

together with the elementary bound in the inner disk $|\zeta| < 2\varepsilon$ (where $\zeta = \varepsilon s$, so that all quadratic forms under consideration are $O(\varepsilon^6)$ after testing against $h = \zeta^3 v$), justifies the expansion in the sense of bounded quadratic forms; the remainder is controlled by

$$C\varepsilon^4\int_{\mathbb{D}}|v|^2\,dA \leq C'\varepsilon^4\int_{\mathbb{D}}|h|^2|\zeta|^{-2}dA.$$

The angular Fourier integrals give

$$\int_0^{2\pi} e^{id\theta}\,d\theta = 2\pi\delta_{d0}, \qquad \int_0^{2\pi}\cos\theta\,e^{i\theta}d\theta = \pi, \qquad \int_0^{2\pi}\big(3\cos^2\theta-1\big)d\theta = \pi,$$

which yield, after the radial integration,

$$D_{ii} = \int_0^1 \rho^{2i}\,\frac{2\pi}{\rho}\,d\rho = \frac{\pi}{i},$$

$$B_{i,i+1} = -\int_0^1 \rho^{2i+1}\,\frac{\pi}{\rho^2}\,d\rho = -\frac{\pi}{2i},$$

$$C_{ii} = \int_0^1 \rho^{2i}\,\frac{\pi}{2\rho^3}\,d\rho = \frac{\pi}{4(i-1)}.$$

The first-order off-diagonal entries of B with $|i-j| \neq 1$ vanish by angular orthogonality.

For the inverse, we use the Neumann expansion

$$A^{-1} = D^{-1}-\varepsilon D^{-1}BD^{-1}+\varepsilon^2\big(D^{-1}BD^{-1}BD^{-1}-D^{-1}CD^{-1}\big)+O(\varepsilon^3).$$

Since D is diagonal, the second-order contribution to the $(3,3)$-entry receives only the diagonal term C_{33} and the unique off-diagonal path $3 \to 4 \to 3$, so that

$$\begin{aligned}\big(A^{-1}\big)_{33} &= \frac{3}{\pi}+\varepsilon^2\Big(\frac{B_{34}^2}{D_{33}^2D_{44}}-\frac{C_{33}}{D_{33}^2}\Big)+O(\varepsilon^4)\\ &= \frac{3}{\pi}+\varepsilon^2\Big(\frac{(\pi/6)^2}{(\pi/3)^2(\pi/4)}-\frac{\pi/8}{(\pi/3)^2}\Big)+O(\varepsilon^4)\\ &= \frac{3}{\pi}-\frac{1}{8\pi}\varepsilon^2+O(\varepsilon^4).\end{aligned}$$

There is no third-order term: the transformation $\zeta \mapsto -\zeta$ unitarily identifies $A(\varepsilon)$ with $A(-\varepsilon)$, while the coefficient functional is unchanged, so the expansion of $\big(A^{-1}\big)_{33}$ contains only even powers. ■

Combining Proposition 2.4 with Proposition 2.5 gives the asymptotic law for the kernel.

Corollary 2.6. *As $q \to \infty$,*

$$K_3(q) = q\big(A(q^{-1})^{-1}\big)_{33} = \frac{3q}{\pi}\Big(1-\frac{1}{24q^2}+O(q^{-4})\Big),$$

and therefore, with $C_0 := \log\big(2(6!)^2\cdot 3/\pi\big)$,

$$\log K_{D_q}^{z^6}(0) = C_0+\log q-\frac{1}{24q^2}+O(q^{-4}).$$

2.4 Proof of the main theorem

Proof of Theorem 1.2. Let $Q > 1$ and $c = 2Q$. By Proposition 2.2, the three values

$$a_- = -\log 2, \qquad a_0 = -\tfrac{1}{2}\log 2, \qquad a_+ = 0$$

correspond to

$$q_- = \frac{Q}{2}, \qquad q_0 = Q, \qquad q_+ = 2Q.$$

Since $a_0 = (a_- + a_+)/2$ and $\log q = \log c + 2a$ is affine in a, it suffices to prove

$$\log K^{z^6}_{D_Q}(0) - \frac{1}{2}\Big(\log K^{z^6}_{D_{Q/2}}(0) + \log K^{z^6}_{D_{2Q}}(0)\Big) > 0$$

for Q sufficiently large. By Corollary 2.6, the terms C_0 cancel, and so do the linear terms $\log q$:

$$\log Q - \frac{1}{2}\big(\log(Q/2) + \log(2Q)\big) = 0.$$

The remaining contribution is

$$-\frac{1}{24}\Big(\frac{1}{Q^2} - \frac{1}{2}\Big(\frac{4}{Q^2} + \frac{1}{4Q^2}\Big)\Big) + O(Q^{-4}) = \frac{3}{64Q^2} + O(Q^{-4}) > 0$$

for Q large enough. This proves the strict midpoint inequality, hence the failure of convexity. ■

Remark 2.7 (A different convexity question). *We recall that Błocki and Zwonek also conjectured in [3] the convexity of the area function $t \mapsto \log \lambda(\{G_{\Omega,w} < t\})$; counterexamples on annuli were found numerically in [1] and proved in full generality in [3]. That conjecture concerns the volume of sublevel sets and is unrelated to Question 1.1, which concerns the Bergman kernel.*

3 A four-connected domain with no point of equality in the higher order Suita conjecture

Fix three pairwise distinct points $c_1, c_2, c_3 \in \mathbb{D}$ and choose $\rho_0 > 0$ so small that the closed disks $\overline{\mathbb{D}(c_j, 2\rho_0)}$ are pairwise disjoint and contained in $\mathbb{D}$. Let

$$B := \big(\tfrac{1}{2}\rho_0, \rho_0\big) \times \big(\tfrac{1}{2}\rho_0, \rho_0\big) \times \big(\tfrac{1}{2}\rho_0, \rho_0\big) \subset \mathbb{R}^3,$$

where $(\frac{1}{2}\rho_0, \rho_0)$ denotes the open interval in $\mathbb{R}$. For $\mathbf{r} = (r_1, r_2, r_3) \in B$ put

$$D_j(\mathbf{r}) := \overline{\mathbb{D}(c_j, r_j)}, \qquad \Gamma_j(\mathbf{r}) := \partial D_j(\mathbf{r}), \quad j = 1, 2, 3, \qquad \Gamma_4 := \partial\mathbb{D},$$

and

$$\Omega(\mathbf{r}) := \mathbb{D} \setminus \Big(D_1(\mathbf{r}) \cup D_2(\mathbf{r}) \cup D_3(\mathbf{r})\Big).$$

Then $\Omega(\mathbf{r})$ is bounded by the four pairwise disjoint analytic Jordan curves $\Gamma_1(\mathbf{r}), \Gamma_2(\mathbf{r}), \Gamma_3(\mathbf{r}), \Gamma_4$ (see Figure 3).

For $j = 1, 2, 3$ let $u_j(\cdot\,; \mathbf{r})$ be the harmonic function on $\Omega(\mathbf{r})$ with boundary values 1 on $\Gamma_j(\mathbf{r})$ and 0 on the other three curves, and set $u_4 := 1 - u_1 - u_2 - u_3$. Basic facts on harmonic measure used below can be found in [6].

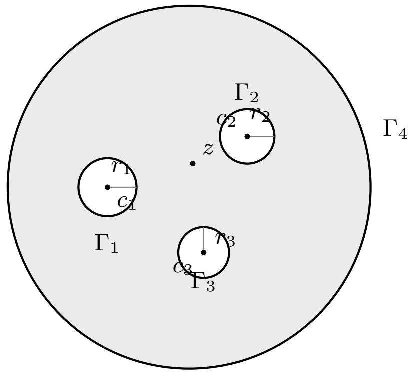

Figure 3: The four-connected domain $\Omega(\mathbf{r}) = \mathbb{D} \setminus \big(D_1(\mathbf{r}) \cup D_2(\mathbf{r}) \cup D_3(\mathbf{r})\big)$ with boundary components $\Gamma_1, \Gamma_2, \Gamma_3$ and Γ_4.

LEMMA 3.1. *For every $z \in \Omega(\mathbf{r})$ and $j, k \in \{1,2,3\}$,*

$$\frac{\partial u_j}{\partial r_j}(z;\mathbf{r}) > 0, \qquad \frac{\partial u_j}{\partial r_k}(z;\mathbf{r}) < 0 \quad (j \neq k), \qquad \frac{\partial u_4}{\partial r_k}(z;\mathbf{r}) < 0. \tag{3.1}$$

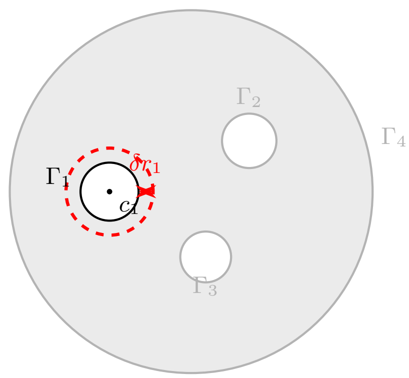

Figure 4: Enlargement of the boundary component $\Gamma_1(\mathbf{r})$ by $\delta r_1 > 0$: the dashed circle is the new boundary, and the red arrow indicates the outward normal displacement used in the variational formula (3.2).

Proof. The functions u_j are real-analytic in $(z, \mathbf{r})$; this follows from the standard analytic dependence of solutions of the Dirichlet problem on analytic boundary data [10]. The first inequality in (3.1) is the usual monotonicity of harmonic measure under enlargement of the corresponding boundary component (Figure 4); it follows from the maximum principle, since on the enlarged domain the old solution is < 1 on the new boundary and $= 0$ on all other boundary components.

For $k \neq j$ (and for u_4 with any k) we use Hadamard's variational formula. Let $G_{\mathbf{r}}(w,z)$ be the Green function of $\Omega(\mathbf{r})$ and let n denote the outward unit normal. Up to a fixed positive multiplicative constant,

$$\frac{\partial u_j}{\partial r_k}(z;\mathbf{r}) = \int_{\Gamma_k(\mathbf{r})} \frac{\partial u_j}{\partial n}(w)\, \frac{\partial G_{\mathbf{r}}}{\partial n}(w,z)\, ds(w). \tag{3.2}$$

For an inner boundary $\Gamma_k(\mathbf{r})$ one has $\partial G_{\mathbf{r}}/\partial n > 0$ on $\Gamma_k(\mathbf{r})$. By Hopf's lemma, $\partial u_j/\partial n$ on $\Gamma_k(\mathbf{r})$ is strictly positive if $j = k$ and strictly negative if $j \neq k$; similarly $\partial u_4/\partial n < 0$ on $\Gamma_k(\mathbf{r})$. Hence (3.1) follows from (3.2). ■

For $z \in \Omega(\mathbf{r})$ put

$$A(z;\mathbf{r}) := \Big(\tfrac{\partial u_j}{\partial r_k}(z;\mathbf{r})\Big)_{1\le j,k\le 3}.$$

Lemma 3.2. *For every $\mathbf{r} \in B$ and every $z \in \Omega(\mathbf{r})$ the matrix $A(z;\mathbf{r})$ is invertible.*

Proof. By (3.1), $A_{jk} < 0$ for $j \neq k$, and

$$(A^T\mathbf{1})_j = \sum_{i=1}^{3} \frac{\partial u_i}{\partial r_j} = \frac{\partial}{\partial r_j}\big(u_1 + u_2 + u_3\big) = -\frac{\partial u_4}{\partial r_j} > 0.$$

Thus A^T has positive diagonal, nonpositive off-diagonal entries and positive row sums. If $A^T\mathbf{x} = 0$ for some $\mathbf{x} \neq 0$, choose an index i at which $x_i = \max_j x_j > 0$; if all $x_j \le 0$, choose instead an index i at which $x_i = \min_j x_j < 0$. In the first case, since $A^T_{ik} \le 0$ for $k \neq i$,

$$0 = (A^T\mathbf{x})_i = A^T_{ii}x_i + \sum_{k\neq i} A^T_{ik}x_k \ge A^T_{ii}x_i + \sum_{k\neq i} A^T_{ik}x_i = (A^T\mathbf{1})_i x_i > 0,$$

a contradiction. The second case is analogous with all inequalities reversed. Hence A^T, and therefore A, is nonsingular. ■

Proof of Theorem 1.3. Let

$$X := \big\{(z,\mathbf{r}) \in \mathbb{C} \times B : z \in \Omega(\mathbf{r})\big\},$$

an open subset of $\mathbb{R}^2 \times \mathbb{R}^3 \cong \mathbb{R}^5$, and define

$$F : X \to \mathbb{R}^3, \qquad F(z,\mathbf{r}) := \big(u_1(z;\mathbf{r}), u_2(z;\mathbf{r}), u_3(z;\mathbf{r})\big).$$

By Lemma 3.1, F is real-analytic. Its derivative contains the invertible 3×3 block $D_{\mathbf{r}}F = A(z;\mathbf{r})$; hence F is a submersion at every point of X.

For $\mathbf{q} = (q_1,q_2,q_3) \in \mathbb{Q}^3$ set $S_{\mathbf{q}} := F^{-1}(\mathbf{q})$. By the implicit function theorem, $S_{\mathbf{q}}$ is a real-analytic submanifold of X of dimension $\dim_{\mathbb{R}} X - 3 = 2$ (possibly empty). Let $\pi_{\mathbf{q}} : S_{\mathbf{q}} \to B$ be the projection onto the parameter box. Each connected component of $S_{\mathbf{q}}$ is a two-dimensional C^1 manifold. Since $\dim_{\mathbb{R}} B = 3$, Sard's theorem shows that its image under $\pi_{\mathbf{q}}$ has empty interior (indeed Lebesgue measure zero). A second countable manifold has at most countably many connected components, so $\pi_{\mathbf{q}}(S_{\mathbf{q}})$ is a countable union of null sets, hence a null set. Consequently

$$E := \bigcup_{\mathbf{q}\in\mathbb{Q}^3} \pi_{\mathbf{q}}(S_{\mathbf{q}})$$

is a countable union of null sets, and $B \setminus E$ is nonempty (in fact of full measure).

Choose $\mathbf{r} \in B \setminus E$. We claim that $\Omega := \Omega(\mathbf{r})$ has the required property. Indeed, if for some $z \in \Omega$ all three numbers $u_j(z;\mathbf{r})$ were rational, then with $\mathbf{q} := \big(u_1(z;\mathbf{r}), u_2(z;\mathbf{r}), u_3(z;\mathbf{r})\big) \in \mathbb{Q}^3$ we would have $(z,\mathbf{r}) \in S_{\mathbf{q}}$ and hence $\mathbf{r} \in E$, a contradiction. This completes the proof. ■

3.1 Generalization to arbitrary connectivity

The construction of the preceding part extends verbatim to every $n \geq 4$.

THEOREM 3.3. *For every integer $n \geq 4$ there exists a planar domain $\Omega \subset \mathbb{C}$ bounded by n analytic Jordan curves such that, for the corresponding harmonic functions $u_1, \ldots, u_{n-1}$, there does not exist $z \in \Omega$ with*

$$u_j(z) \in \mathbb{Q}, \qquad j = 1, \ldots, n-1.$$

In particular, at no point of Ω does the equality in the higher order Suita conjecture hold.

Proof. Fix $n \geq 4$ and set $m := n-1$. Choose m pairwise distinct points $c_1, \ldots, c_m \in \mathbb{D}$ and $\rho_0 > 0$ so small that the closed disks $\overline{\mathbb{D}(c_j, 2\rho_0)}$ are pairwise disjoint and contained in $\mathbb{D}$. Let $B := (\rho_0/2, \rho_0)^m$. For $\mathbf{r} = (r_1, \ldots, r_m) \in B$ define

$$D_j(\mathbf{r}) := \overline{\mathbb{D}(c_j, r_j)}, \qquad \Gamma_j(\mathbf{r}) := \partial D_j(\mathbf{r}) \quad (1 \leq j \leq m), \qquad \Gamma_{m+1} := \partial \mathbb{D},$$

and

$$\Omega(\mathbf{r}) := \mathbb{D} \setminus \bigcup_{j=1}^{m} D_j(\mathbf{r}).$$

Let $u_j(\cdot\,; \mathbf{r})$ be the harmonic function on $\Omega(\mathbf{r})$ with boundary values 1 on $\Gamma_j(\mathbf{r})$ and 0 on the other curves, $1 \leq j \leq m$, and put $u_{m+1} := 1 - \sum_{j=1}^{m} u_j$.

The variational argument in Lemma 3.1 applies unchanged with m in place of 3: for all $j, k \in \{1, \ldots, m\}$,

$$\frac{\partial u_j}{\partial r_j}(z; \mathbf{r}) > 0, \qquad \frac{\partial u_j}{\partial r_k}(z; \mathbf{r}) < 0 \quad (j \neq k), \qquad \frac{\partial u_{m+1}}{\partial r_k}(z; \mathbf{r}) < 0.$$

Indeed, the off-diagonal and u_{m+1} estimates follow from Hadamard's formula (3.2) and Hopf's lemma exactly as in the proof of Lemma 3.1.

Let $A(z; \mathbf{r}) := (\partial u_j / \partial r_k)_{1 \leq j,k \leq m}$. As in Lemma 3.2,

$$(A^T \mathbf{1})_j = \sum_{i=1}^{m} \frac{\partial u_i}{\partial r_j} = \frac{\partial}{\partial r_j}\Big(\sum_{i=1}^{m} u_i\Big) = -\frac{\partial u_{m+1}}{\partial r_j} > 0.$$

Thus A^T has positive diagonal, nonpositive off-diagonal entries and positive row sums; by the same maximum-entry argument as in Lemma 3.2, A^T, and hence A, is nonsingular.

Define

$$X := \big\{(z, \mathbf{r}) \in \mathbb{C} \times B : z \in \Omega(\mathbf{r})\big\}, \qquad F : X \to \mathbb{R}^m, \quad F(z, \mathbf{r}) := \big(u_1(z; \mathbf{r}), \ldots, u_m(z; \mathbf{r})\big).$$

Then F is real-analytic and a submersion, since $D_{\mathbf{r}} F = A$ is invertible. For every $\mathbf{q} \in \mathbb{Q}^m$, the set $S_{\mathbf{q}} := F^{-1}(\mathbf{q})$ is a real-analytic submanifold of X of dimension

$$\dim_{\mathbb{R}} X - m = (2 + m) - m = 2.$$

Let $\pi_{\mathbf{q}} : S_{\mathbf{q}} \to B$ be the projection onto the parameter box. Since $\dim_{\mathbb{R}} B = m = n - 1 \geq 3$, Sard's theorem shows that $\pi_{\mathbf{q}}(S_{\mathbf{q}})$ is a null set: every connected component is a smooth image of a two-dimensional manifold, and there are at most countably many components. Therefore

$$E := \bigcup_{\mathbf{q} \in \mathbb{Q}^m} \pi_{\mathbf{q}}(S_{\mathbf{q}})$$

is a countable union of null sets, and $B \setminus E$ is nonempty.

Choose $\mathbf{r} \in B \setminus E$. If for some $z \in \Omega(\mathbf{r})$ all $u_j(z; \mathbf{r})$ were rational, then $(z, \mathbf{r}) \in S_{\mathbf{q}}$ for $\mathbf{q} := F(z, \mathbf{r}) \in \mathbb{Q}^m$, so $\mathbf{r} \in E$, a contradiction. Hence $\Omega := \Omega(\mathbf{r})$ satisfies the conclusion of the theorem. ■

4 The ξ-reduction fails for $0 < p < 1$

Let $D \subset \mathbb{C}^n$ be a domain, $p > 0$, and let

$$H(z) = \sum_{|\alpha|=k} a_\alpha z^\alpha$$

be a homogeneous polynomial of degree k. Following [9], denote by

$$\ell_1^{(n)} := \Big\{ \xi = (\xi_\alpha)_{\alpha \in \mathbb{N}^n} : \ \xi_\alpha \in \mathbb{C}, \ \sum_{\alpha \in \mathbb{N}^n} |\xi_\alpha| \rho^{|\alpha|} < +\infty \text{ for every } \rho > 0 \Big\}$$

the space of functionals on holomorphic germs. For $\xi \in \ell_1^{(n)}$ and $f \in \mathcal{O}(D)$, the pairing is defined by

$$(\xi \cdot f)(z) := \sum_{\alpha \in \mathbb{N}^n} \xi_\alpha \frac{D^\alpha f(z)}{\alpha!}.$$

For the homogeneous polynomial H of degree k above, let S_H denote the affine family of all $\xi \in \ell_1^{(n)}$ such that

$$\xi_\alpha = \begin{cases} a_\alpha \alpha!, & |\alpha| = k, \\ 0, & |\alpha| > k, \end{cases}$$

the entries with $|\alpha| < k$ being free. With this notation, the <u>higher order p-Bergman kernel</u> and the <u>L^p ξ-Bergman kernel</u> are

$$K_D^{H,p}(z) := \sup\Big\{ |P_H(f)(z)|^p : \ f \in A^p(D), \ f^{(j)}(z) = 0 \ (0 \le j < k), \ \|f\|_{L^p(D)} \le 1 \Big\},$$

$$K_{\xi,D,p}(z) := \sup\Big\{ |(\xi \cdot f)(z)|^p : \ f \in A^p(D), \ \|f\|_{L^p(D)} \le 1 \Big\}.$$

For a positive homogeneous functional, equivalently,

$$K_{\xi,D,p}(z) = \frac{1}{\inf\big\{ \|f\|_{L^p(D)}^p : \ f \in A^p(D), \ (\xi \cdot f)(z) = 1 \big\}}.$$

We answer the question whether the identity in Theorem 1.7 of [9] extends to $0 < p < 1$ in the negative. Already the one-dimensional model $D = \mathbb{D}$, $H(z) = z$, $p = 1/2$ gives a counterexample. For $H(z) = z$ the set S_H consists of all sequences

$$\xi = (\xi_0, 1, 0, 0, \ldots), \qquad \xi_0 \in \mathbb{C}.$$

Proof of Theorem 1.4. We first compute $K_{\mathbb{D}}^{H,1/2}(0)$. Let $f \in \mathcal{O}(\mathbb{D})$ satisfy $f(0) = 0$ and $f'(0) = 1$, and write $f(z) = zg(z)$ with $g(0) = 1$. Since $|g|^{1/2}$ is subharmonic, for every $r \in (0, 1)$,

$$\frac{1}{2\pi}\int_0^{2\pi} |g(re^{i\theta})|^{1/2}\, d\theta \geq |g(0)|^{1/2} = 1.$$

Hence

$$\int_{\mathbb{D}} |f|^{1/2}\, dA = \int_0^1 r^{1/2} r\Big(\int_0^{2\pi} |g(re^{i\theta})|^{1/2}\, d\theta\Big) dr \geq 2\pi \int_0^1 r^{3/2}\, dr = \frac{4\pi}{5},$$

with equality for $g \equiv 1$, i.e. $f(z) = z$. Therefore

$$K_{\mathbb{D}}^{H,1/2}(0) = \frac{1}{4\pi/5} = \frac{5}{4\pi}.$$

Next, fix $\alpha = 3/2$ and set

$$M(\alpha) := \frac{1}{\pi}\int_{\mathbb{D}} |1 + \alpha z|\, dA(z).$$

We need the elementary estimate

$$\frac{M(\alpha)}{\sqrt{2\alpha}} < \frac{4}{5}. \tag{4.3}$$

For $0 \leq x \leq 1$ the fourth Taylor polynomial of $\sqrt{1 - x}$ gives

$$\sqrt{1 - x} \leq 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} - \frac{5x^4}{128}, \tag{4.4}$$

because all remaining Taylor coefficients of $\sqrt{1 - x}$ are negative. Write $t = \alpha r$ and

$$\kappa := \frac{4t}{(1 + t)^2}.$$

Then

$$|1 + te^{i\theta}| = (1 + t)\sqrt{1 - \kappa \sin^2 \frac{\theta}{2}}.$$

Using (4.4) and the averages

$$\frac{1}{2\pi}\int_0^{2\pi} \sin^2\frac{\theta}{2}\, d\theta = \frac{1}{2}, \quad \frac{1}{2\pi}\int_0^{2\pi} \sin^4\frac{\theta}{2}\, d\theta = \frac{3}{8}, \quad \frac{1}{2\pi}\int_0^{2\pi} \sin^6\frac{\theta}{2}\, d\theta = \frac{5}{16}, \quad \frac{1}{2\pi}\int_0^{2\pi} \sin^8\frac{\theta}{2}\, d\theta = \frac{35}{128},$$

we obtain

$$\frac{1}{2\pi}\int_0^{2\pi} |1 + te^{i\theta}|\, d\theta \leq (1 + t)\Big(1 - \frac{\kappa}{4} - \frac{3\kappa^2}{64} - \frac{5\kappa^3}{256} - \frac{175\kappa^4}{16384}\Big).$$

Consequently

$$M(\alpha) \leq \int_0^1 2r(1 + \alpha r)\Big(1 - \frac{\kappa}{4} - \frac{3\kappa^2}{64} - \frac{5\kappa^3}{256} - \frac{175\kappa^4}{16384}\Big)\, dr.$$

For $\alpha = 3/2$ the integral on the right is

$$\frac{13771}{10000}.$$

Since
$$\left(\frac{13771}{10000}\right)^2 = \frac{189640441}{100000000} < \frac{48}{25} = \left(\frac{4}{5}\sqrt{3}\right)^2,$$
we have
$$\frac{13771}{10000} < \frac{4}{5}\sqrt{3} = \frac{4}{5}\sqrt{2\alpha},$$
which proves (4.3).

Now fix any $c \in \mathbb{C}$. Choose a unit complex number $e^{i\theta}$ such that c and $2\alpha e^{i\theta}$ have the same argument (take $e^{i\theta} = 1$ if $c = 0$), so that $c + 2\alpha e^{i\theta}$ has modulus $|c| + 2\alpha$. Choose a unit complex number $e^{i\psi}$ with $e^{i\psi}(c + 2\alpha e^{i\theta}) = |c| + 2\alpha$, and define
$$f_{c,\theta}(z) := e^{i\psi} \frac{(1 + \alpha e^{i\theta} z)^2}{|c| + 2\alpha}.$$
Then $f_{c,\theta}$ is a polynomial, and
$$f_{c,\theta}(0) = \frac{e^{i\psi}}{|c| + 2\alpha}, \qquad f'_{c,\theta}(0) = \frac{2\alpha e^{i\theta} e^{i\psi}}{|c| + 2\alpha},$$
so that for $\xi_c = (c, 1, 0, 0, \ldots) \in S_H$,
$$(\xi_c \cdot f_{c,\theta})(0) = c f_{c,\theta}(0) + f'_{c,\theta}(0) = \frac{e^{i\psi}(c + 2\alpha e^{i\theta})}{|c| + 2\alpha} = 1.$$
Moreover, by the rotation invariance of dA,
$$\int_{\mathbb{D}} |f_{c,\theta}|^{1/2}\, dA = \frac{1}{\sqrt{|c| + 2\alpha}} \int_{\mathbb{D}} |1 + \alpha e^{i\theta} z|\, dA = \frac{\pi M(\alpha)}{\sqrt{|c| + 2\alpha}} \le \frac{\pi M(\alpha)}{\sqrt{2\alpha}} < \frac{4\pi}{5},$$
since $|c| + 2\alpha \ge 2\alpha$. By the equivalent form of $K_{\xi,D,p}$,
$$K_{\xi_c,\mathbb{D},1/2}(0) \ge \frac{1}{\int_{\mathbb{D}} |f_{c,\theta}|^{1/2}\, dA} > \frac{5}{4\pi} = K_{\mathbb{D}}^{H,1/2}(0).$$
This holds for every $c \in \mathbb{C}$, i.e. for every $\xi \in S_H$.

Thus for every $\xi \in S_H$,
$$K_{\xi,\mathbb{D},1/2}(0) > \frac{5}{4\pi},$$
and hence
$$\inf_{\xi \in S_H} K_{\xi,\mathbb{D},1/2}(0) > \frac{5}{4\pi} = K_{\mathbb{D}}^{H,1/2}(0),$$
which proves the theorem. ■

4.1 The threshold $p_* = 2(\sqrt{2} - 1)$ for the counterexample

The counterexample in Theorem 1.4 is not an isolated phenomenon: the second variation at the extremal $f(z) = z$ shows that the identity
$$K_{\mathbb{D}}^{H,p}(0) = \inf_{\xi \in S_H} K_{\xi,\mathbb{D},p}(0) \tag{4.5}$$

fails for every $0 < p < p_*$, where

$$p_* := 2(\sqrt{2} - 1).$$

For $p > p_*$ the same computation shows that z is a strict local minimizer in the class $f'(0) = 1$, and at $p = p_*$ the second variation degenerates along a single explicit direction; numerically, z remains the global minimizer for every $p \geq p_*$ in this example.

Since $H(z) = z$, every $\xi \in S_H$ has the form $\xi_c = (c, 1, 0, 0, \ldots)$, $c \in \mathbb{C}$, and

$$K_{\xi_c,\mathbb{D},p}(0) = \frac{1}{m_{\xi_c}(p)}, \qquad m_{\xi_c}(p) := \inf\Big\{ \int_{\mathbb{D}} |f|^p \, dA : \ f \in \mathcal{O}(\mathbb{D}), \ cf(0) + f'(0) = 1 \Big\}.$$

The function $f(z) = z$ is admissible for every ξ_c, and $\int_{\mathbb{D}} |z|^p \, dA = 2\pi/(p+2)$; hence $m_{\xi_c}(p) \leq 2\pi/(p+2)$ for all c.

THEOREM 4.1. *Let $p_* := 2(\sqrt{2} - 1)$. For every $0 < p < p_*$, with $D = \mathbb{D}$, $H(z) = z$ and $z = 0$,*

$$\inf_{\xi \in S_H} K_{\xi,\mathbb{D},p}(0) > K^{H,p}_{\mathbb{D}}(0);$$

that is, the identity (4.5) fails for every $p \in (0, p_*)$. In particular, Theorem 4.1 contains the case $p = 1/2$ of Theorem 1.4.

Proof. By the same argument as in the proof of Theorem 1.4, with p in place of $1/2$,

$$K^{H,p}_{\mathbb{D}}(0) = \frac{p+2}{2\pi}.$$

It suffices to show that $m_{\xi_0}(p) < 2\pi/(p+2)$. For $\lambda > 0$ and $\varepsilon \in \mathbb{R}$ consider

$$f_\varepsilon(z) := z + \varepsilon(1 + \lambda z^2),$$

so that $f'_\varepsilon(0) = 1$ and $f_0 = z$. Write $J_p(\varepsilon) := \int_{\mathbb{D}} |f_\varepsilon|^p \, dA$. The first derivative at 0 vanishes by symmetry, and a direct expansion (given after the proof) yields

$$J''_p(0) = \pi\Big(p + \frac{p^2\lambda^2}{p+4} + \frac{2p(p-2)\lambda}{p+2} \Big). \tag{4.6}$$

The quadratic polynomial in λ on the right-hand side has minimum

$$p - \frac{(2-p)^2(p+4)}{(p+2)^2} = \frac{4(p^2+4p-4)}{(p+2)^2} = \frac{4(p-p_*)(p+2+2\sqrt{2})}{(p+2)^2},$$

attained at

$$\lambda_p := \frac{(2-p)(p+4)}{p(p+2)}.$$

Since $p + 2 + 2\sqrt{2} > 0$ for $p > 0$, the minimum is negative precisely for $p < p_*$. Fixing such a p and taking $\lambda = \lambda_p$, we have $J''_p(0) < 0$; as $J'_p(0) = 0$ and $J_p(0) = 2\pi/(p+2)$, for all sufficiently small $\varepsilon > 0$,

$$\int_{\mathbb{D}} |f_\varepsilon|^p \, dA < \frac{2\pi}{p+2}.$$

Thus $m_{\xi_0}(p) < 2\pi/(p+2)$, and Proposition 4.2 yields $\sup_{c\in\mathbb{C}} m_{\xi_c}(p) < 2\pi/(p+2)$. Consequently

$$\inf_{\xi\in S_H} K_{\xi,\mathbb{D},p}(0) = \frac{1}{\sup_{c\in\mathbb{C}} m_{\xi_c}(p)} > \frac{p+2}{2\pi} = K^{H,p}_{\mathbb{D}}(0),$$

■

Proposition 4.2. *The identity* (4.5) *holds if and only if* $m_{\xi_0}(p) = 2\pi/(p+2)$, *i.e. if and only if* z *minimizes* $\int_{\mathbb{D}} |f|^p \, dA$ *among all holomorphic* f *with* $f'(0) = 1$.

Proof. Since

$$\inf_{\xi \in S_H} K_{\xi,\mathbb{D},p}(0) = \inf_{c \in \mathbb{C}} \frac{1}{m_{\xi_c}(p)} = \frac{1}{\sup_{c\in\mathbb{C}} m_{\xi_c}(p)},$$

it suffices to show that $\sup_{c\in\mathbb{C}} m_{\xi_c}(p)$ equals $2\pi/(p+2)$ if and only if it is attained at $c = 0$.

We first note that $c \mapsto m_{\xi_c}(p)$ is continuous. Write $J_p(f) := \int_{\mathbb{D}} |f|^p \, dA$ and

$$N(c) := \sup_{f \neq 0} \frac{|cf(0) + f'(0)|}{J_p(f)^{1/p}},$$

so that $m_{\xi_c}(p) = N(c)^{-p}$. Since $|f(0)| \leq \pi^{-1/p} J_p(f)^{1/p}$ by subharmonicity,

$$|N(c) - N(d)| \leq |c - d| \, \pi^{-1/p},$$

so N, and hence $c \mapsto m_{\xi_c}(p)$, is continuous. Taking $f \equiv 1$ shows $N(c) \geq |c| \, \pi^{-1/p}$, whence $m_{\xi_c}(p) \leq \pi \, |c|^{-p} \to 0$ as $|c| \to \infty$.

It remains to check that $m_{\xi_c}(p) < 2\pi/(p+2)$ for every $c \neq 0$. Set $\eta = \bar{c}/|c|$ and $f_t(z) = z + t\eta(1 - cz)$. Then $cf_t(0) + f_t'(0) = 1$ for every t, and

$$\frac{d}{dt}\Big|_{t=0} \int_{\mathbb{D}} |f_t|^p \, dA = p \, \Re \int_{\mathbb{D}} |z|^{p-2} \bar{z} \, \eta(1 - cz) \, dA = -p \, \Re(\eta c) \int_{\mathbb{D}} |z|^p \, dA = -p|c| \, \frac{2\pi}{p+2} < 0,$$

because $\int_{\mathbb{D}} |z|^{p-2} \bar{z} \, dA = 0$ by rotation invariance. Thus $J_p(f_t) < 2\pi/(p+2)$ for all sufficiently small $t > 0$, which gives $m_{\xi_c}(p) < 2\pi/(p+2)$.

Consequently, if $m_{\xi_0}(p) = 2\pi/(p+2)$ then the supremum is attained at $c = 0$; if instead $m_{\xi_0}(p) < 2\pi/(p+2)$, then $m_{\xi_c}(p) < 2\pi/(p+2)$ for every $c \in \mathbb{C}$, and continuity together with the decay at infinity yields $\sup_{c\in\mathbb{C}} m_{\xi_c}(p) < 2\pi/(p+2)$. ∎

For completeness we record the second variation used above. Let $h(z) = \sum_{m \neq 1} h_m z^m$ be holomorphic with $h'(0) = 0$. Expanding $|z + \varepsilon h|^p = |z|^p \, |1 + \varepsilon h/z|^p$ pointwise for $z \neq 0$ and using the elementary averages $\frac{1}{2\pi} \int_0^{2\pi} e^{ik\theta} \, d\theta = \delta_{k,0}$, we obtain

$$\frac{d^2}{d\varepsilon^2}\Big|_{\varepsilon=0} \int_{\mathbb{D}} |z + \varepsilon h|^p \, dA = \pi \left(p^2 \sum_{m \neq 1} \frac{|h_m|^2}{p + 2m} + \frac{2p(p-2)}{p+2} \Re(h_0 h_2) \right), \tag{4.7}$$

which for $h(z) = 1 + \lambda z^2$ reduces to (4.6). The only potentially negative term is the cross term $\Re(h_0 h_2)$, which arises from the pairing of the modes $h_0 z^{-1}$ and $h_2 z$ in $(h/z)^2$. The threshold p_* is exactly the value at which the two-mode quadratic form $(x, y) \mapsto px^2 + p^2y^2/(p+4) + 2p(p-2)xy/(p+2)$ becomes positive semidefinite.

Remark 4.3. *It remains open whether* (4.5) *holds for every* $p \geq p_*$. *For* $p \geq 1$ *this follows from [9, Theorem 1.7], and the numerical evidence above suggests that* z *remains the global minimizer for every* $p \geq p_*$ *in this example; a rigorous proof of* (4.5) *in the remaining range* $p_* \leq p < 1$ *is not known.*

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